Innovation Diffusion Analysis Using the Bass Model

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Course: Homework on Bass Model
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1. Introduction

For this project, I chose the Unitree R1 humanoid robot, which was included in TIME's Best Inventions of 2025. I wanted to study how a new technology like this could spread in the market over time.

To do this, I used the Bass Model, which helps explain how people and businesses start adopting new products. Since there is not enough historical data about Unitree R1 sales, I used worldwide industrial robot shipment data from previous years to estimate the model.

Industrial robots and humanoid robots are not exactly the same, so the results cannot tell us exactly how many Unitree R1 robots will be sold in the future. Instead, I used the model to understand possible patterns of growth and see how different market assumptions affect the results.

Selected innovation: Unitree R1 humanoid robot

TIME article: https://time.com/collections/best-inventions-2025/7318495/unitree-r1/

2. Comparison with a Previous Innovation

I chose traditional industrial robots as a comparison because they are an older technology that has already been used in many industries. Both industrial and humanoid robots use sensors, software, and mechanical systems to perform different tasks.

Industrial robots have been used in manufacturing for many years, so their shipment data can help us understand how the robotics market has developed.

However, there are also some important differences. Traditional industrial robots are usually designed to perform specific tasks in factories, while the Unitree R1 is a humanoid robot that can be used for research, education, and development.

The customers and uses of these robots are not necessarily the same. Because of this, I used industrial robot data as a reference for the analysis, rather than assuming that humanoid robots will follow exactly the same pattern.

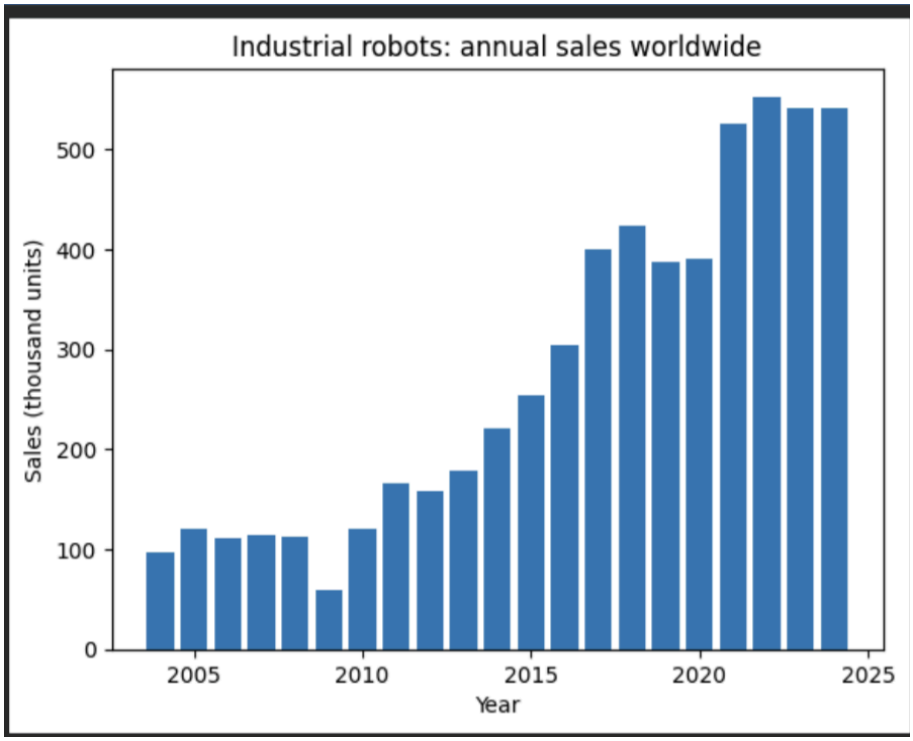
3. Data and Geographical Scope

For the historical part of the project, I used worldwide industrial robot shipment data from 2004 to 2024. The data are measured in thousands of units and are attributed to the International Federation of Robotics (IFR), through Statista.

Data source:
https://de.statista.com/statistik/daten/studie/29386/umfrage/weltweiter-absatz-fuer-industrie-roboter-seit-2004/

I chose a global scope because the robotics market is international, and the available historical data cover shipments worldwide. I did not focus on one country because I did not have a comparable historical dataset for Unitree R1 adoption in a specific country.

The dataset helped me estimate the Bass Model using an existing technology. I then used the estimated parameters to explore possible future scenarios for humanoid robots.



4. Bass Model Methodology

The Bass Model is used to study how a new product spreads in the market over time. It has three main parameters:

- **p — coefficient of innovation:** This represents adoption that happens because of external influences, such as advertising, information, and general awareness of a product.
- **q — coefficient of imitation:** This represents adoption influenced by people or businesses that have already adopted the product.
- **M — market potential:** This is the total potential market assumed by the model.

I fitted the Bass Model to the historical industrial robot shipment data to estimate these parameters. After that, I calculated the predicted annual shipments by looking at the difference between the model's cumulative predictions for consecutive years.

In this project, I used shipments as an indication of adoption because this was the information available in the dataset. However, shipments do not necessarily tell us how many different customers bought robots or how many robots are still being used.

After estimating the model with the industrial robot data, I used its parameters to create different scenarios for the future of humanoid robots.

5. Bass Model Estimation Results

After fitting the model to the historical industrial robot data, I obtained the following results:

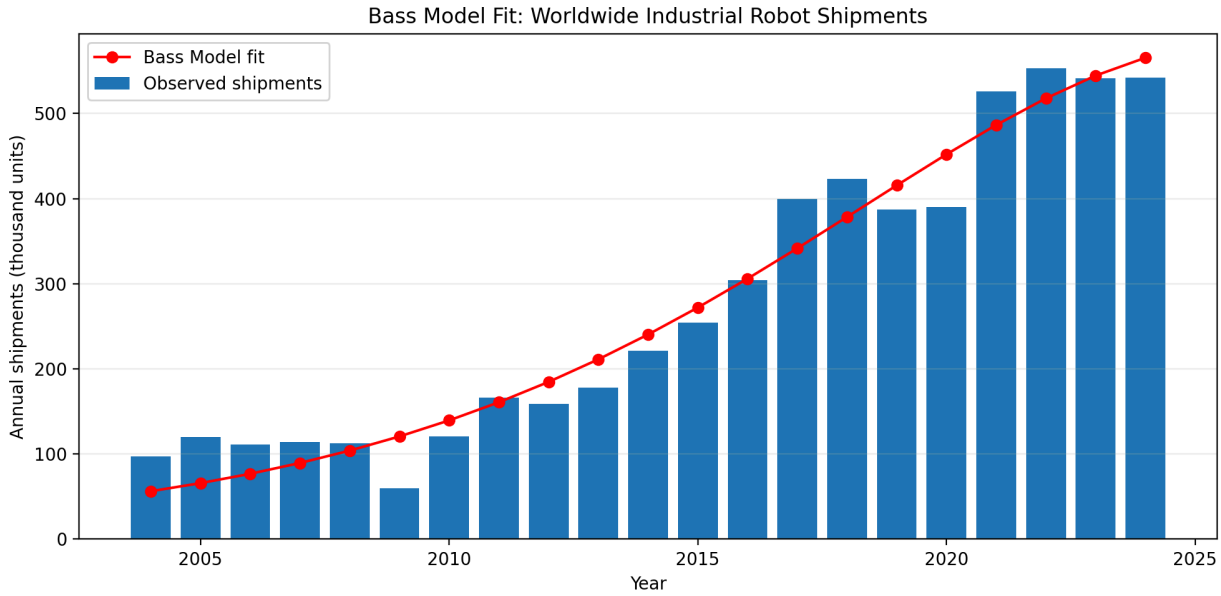
Coefficient of innovation (p)	0.003735
Coefficient of imitation (q)	0.161122
Market potential (M)	13,917.22 thousand units
R-squared	0.9552
RMSE	35.45 thousand units

The estimated market potential is approximately 13.92 million units. This value refers to the industrial robot data used in the model, not to the potential market for Unitree R1.

One thing I noticed is that the imitation coefficient is higher than the innovation coefficient. This means that, according to the fitted model, imitation has a stronger role than external influences in explaining the diffusion pattern.

The R-squared value of 0.9552 suggests that the model fits the general pattern of the historical data quite well. The RMSE is about 35,450 units, which gives an idea of the typical size of the model's prediction errors.

Still, the model does not capture every change in annual shipments. Even if it fits the historical data well, this does not automatically mean that its future predictions will be accurate.



6. Humanoid Robot Diffusion Scenarios

Since I did not have enough historical sales data for Unitree R1, I created three hypothetical scenarios to see how the market could develop under different assumptions.

I used three possible market sizes:

- 10 million units
- 30 million units
- 100 million units

These values are assumptions for the scenario analysis. They are not confirmed estimates of how many humanoid robots will eventually be sold, and they do not represent Unitree R1 sales specifically.

For the starting point, I used an annual shipment estimate of 20,000 humanoid robots for 2025, which I attributed to the Bank of America Institute report *Physical AI, Part 2: Humanoid Robots*. This figure refers to the humanoid robot market in general, not just the Unitree R1.

I then used the imitation coefficient estimated from the historical industrial robot data. For each market-size scenario, I adjusted the innovation coefficient so that the model matched the assumed starting value of 20,000 shipments in 2025.

This allowed me to compare how the model behaves when the potential market is smaller or larger. The three scenarios help show how much the long-term results depend on the assumptions we choose.

However, these projections should not be treated as exact forecasts. The humanoid robot market is still developing, and its growth may be different from the pattern observed for traditional industrial robots. Therefore, the scenarios are mainly useful for exploring possible outcomes rather than predicting actual Unitree R1 sales.

7. Predictions by Period

Using the historical industrial robot data, the Bass Model predicts around 580,000 shipments in 2025 and 536,000 in 2030. By 2030, the cumulative number of shipments is estimated to reach approximately 9.15 million units.

For humanoid robots, I created three scenarios based on different market potentials: 10 million, 30 million, and 100 million units. In all three scenarios, I used 20,000 shipments in 2025 as the starting point, based on a forecast published by the Bank of America Institute. Since there is not enough historical sales data for the Unitree R1, these scenarios help explore how the market could develop in the coming years.

The results are shown below.

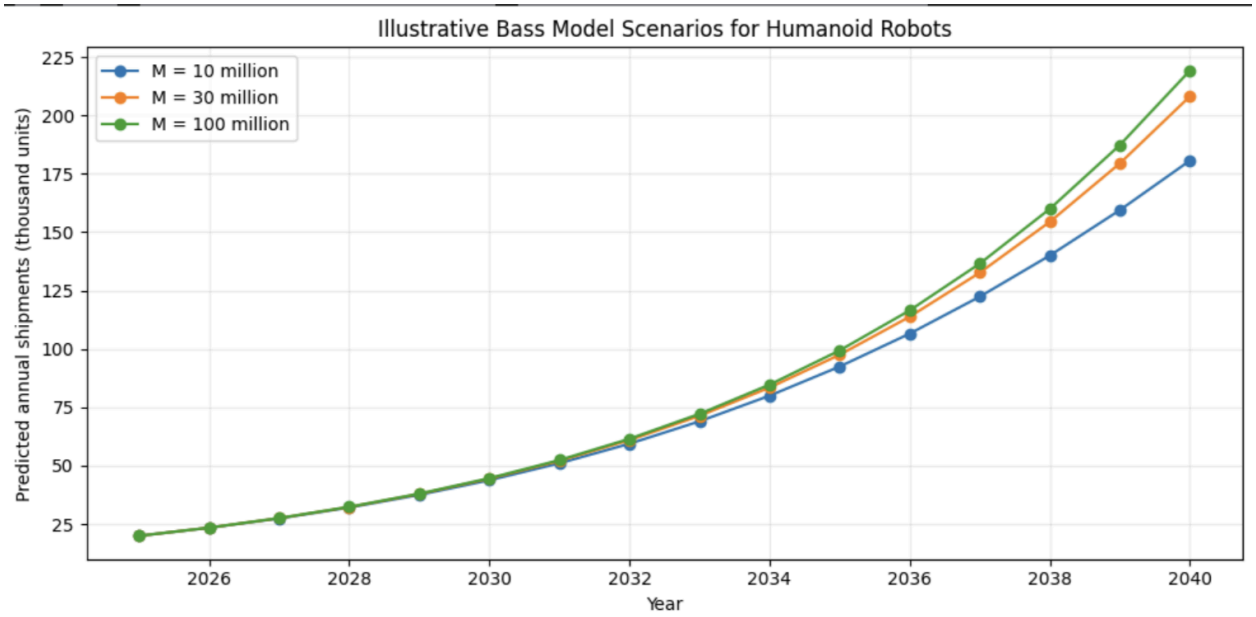
Year	Annual shipments	Cumulative shipments
2025	20,000	20,000

2030	43,808–44,665	184,329–186,185
2040	180,665–219,210	1,245,428–1,376,037

The results show that the model predicts an increase in annual humanoid robot shipments over time. The forecasts are fairly similar across the three scenarios in the earlier years, but the differences become larger over time as the assumed market potential changes.

However, these estimates are much lower than the Bank of America Institute's forecast, which expects annual humanoid robot shipments to reach 1.2 million units by 2030 and 10 million by 2035. This suggests that the results depend heavily on the assumptions used in the model.

It is also important to remember that these predictions are for the humanoid robot market in general, not specifically for the Unitree R1. The cumulative figures represent total shipments over time, not the number of individual customers or robots still in use. Therefore, the results should be treated as possible scenarios rather than exact predictions of future sales.



8. Limitations

There are several limitations to this project that should be considered when looking at the results.

First, industrial robots and humanoid robots are different products. Their uses, customers, prices, and adoption processes may differ, so the historical industrial robot data may not fully represent the future humanoid robot market.

Second, shipment numbers do not necessarily show the number of unique customers or the number of robots currently in use.

Third, the Bass Model does not directly account for factors such as changes in technology, prices, production capacity, regulations, or economic conditions. These factors could affect how quickly humanoid robots are adopted.

Finally, the humanoid robot predictions depend on the assumptions used in the scenarios, including the starting shipment value, imitation coefficient, and market potential. Changing these assumptions could lead to different results.

For these reasons, I consider the results a way to explore possible diffusion patterns rather than precise predictions of future sales.

9. Conclusion

In this project, I used the Bass Model to study the historical development of industrial robot shipments and explore possible future diffusion patterns for humanoid robots.

The results showed that the fitted industrial robot model gives a relatively good representation of the historical data. I then used its estimated parameters to create three scenarios for humanoid robots with different assumed market sizes.

The scenario analysis showed how the market-potential assumption can affect the model's long-term predictions. However, because industrial robots and humanoid robots are different, the results cannot be directly used to predict Unitree R1 sales.

Overall, this project helped me understand how the Bass Model can be used to study the diffusion of an innovation, even when there is not enough historical data for the specific product. For a more reliable analysis, it would be useful to have more years of actual humanoid robot shipment data and information specifically about Unitree R1 sales.

References

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2. Statista. (2025). *Sales of industrial robots worldwide from 2004 to 2024 and forecast to 2028*. Source: International Federation of Robotics. https://www.statista.com/statistics/264084/worldwide-sales-of-industrial-robots/

3. Bank of America Institute. (2026, March 12). *Physical AI, Part 2: Humanoid Robots*. https://institute.bankofamerica.com/content/dam/transformation/physical-ai